Nitesh Kumar

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Professional Summary

GenAI Engineer with 5+ years of experience building enterprise AI applications, RAG systems, intelligent document processing pipelines, and agentic AI solutions. Expertise in Python, LangChain, LangGraph, FastAPI, MongoDB, Google Cloud Platform, prompt engineering, and AI observability. Proven track record delivering production-grade AI solutions for regulatory document processing, conversational AI, and workflow automation.

Skills

Programming Languages: Python, SQL, JSON, Basic PySpark
Generative AI: Gemini 2.5 Pro, Gemini 2.5 Flash, LangChain, LangGraph, RAG, Agentic AI, ReAct Agents, Prompt Engineering, Embeddings, Semantic Search, Vector Search, FAISS
Backend & APIs: FastAPI, AsyncIO, REST APIs, Flask, Postman, Pydantic, HTTPX
System Design: Logical & Physical Architecture, DB Schema Design, Microservices, Scalable API Design, Stateless Services, Multi-user Session Architecture
Databases: MongoDB, SQLite, MySQL
Cloud (GCP): Cloud Run, Google Cloud Storage (GCS), BigQuery, Vertex AI, IAM
Document Processing: BeautifulSoup, PyMuPDF, pdfplumber, ReportLab, Unstructured
Observability & Security: OpenTelemetry, Arize Phoenix, HashiCorp Vault
DevOps: Docker, Kubernetes, Git, Bitbucket, Grafana

Professional Experience

Tata Consultancy Services (TCS) Aug 2024 – Present
GenAI Engineer

Email Ingestion & Processing Pipeline

- Built async email ingestion pipeline using Microsoft Graph API, MSAL OAuth2, and AsyncIO, processing 1,000+ emails and attachments daily with parallel concurrency.
- Developed HTML email parsing (BeautifulSoup) and PDF archival (ReportLab) supporting 5+ attachment types (PDF, DOCX, Excel, images, URLs).
- Enforced SHA-256 deduplication eliminating 100% duplicate records; DLQ and retry mechanisms achieved 99.9% processing reliability.
- Designed 4 MongoDB collections (ingestion tracking, deduplication, DLQ, chunk storage) storing metadata for 10,000+ processed documents in GCS.

State Enterprise Regulatory Processing Pipeline

- Designed 6-stage LangGraph pipeline (normalization, relevancy filtering, LOB classification, metadata extraction, summarization, intake transform) with full checkpointing.
- Architected multi-source ingestion across 5 channels (Bloomberg, Fluxguard, NILS, Email, manual uploads), normalizing 500+ regulatory documents per run into GCS.
- Implemented conditional routing for 3 LOB-specific pipelines (State Enterprise, Medicare, Medicaid), reducing misclassification rate by 90%.
- Enabled pipeline resumability via LangGraph checkpointing, reducing reprocessing cost by 80% on failure recovery.

AI-Powered Document Summarization Pipeline

- Engineered RAG-based summarization pipeline using Gemini LLM and KBS, processing documents of 100+ pages with chunk-level aggregation.
- Reduced manual regulatory summarization time from 4+ hours to under 2 minutes per document using automated AI-driven workflows.
- Developed structured prompt engineering framework producing JSON and PDF outputs with 95%+ metadata extraction accuracy.
- Integrated KBS semantic search fetching top-10 relevant regulatory chunks per query, achieving 100% schema-compliant structured outputs via BusinessRules validation.

AI-Powered Regulatory Chatbot (REGIS)

- Built enterprise chatbot using LangChain ReAct Agents and Gemini 2.5 Pro, reducing regulatory research time from hours to under 30 seconds per query.
- Designed multi-user session architecture supporting 15–20 concurrent users with full session isolation and handling 25,000+ monthly chatbot messages.
- Integrated KBS tool-based retrieval (LOB search, citation lookup, timeline queries) achieving citation-backed responses for 100% of regulatory answers.
- Deployed on Cloud Run with OpenTelemetry and Arize Phoenix tracing across 5+ LLM metrics; secrets managed via HashiCorp Vault.

System & Database Design

- Contributed to logical and physical architecture design for 3+ enterprise AI systems covering React UI, FastAPI backend, AI agent layer, GCS storage, and Cloud Run deployment.
- Designed MongoDB schemas for 8+ collections across chat sessions, audit logs, pipeline state, deduplication, and document metadata supporting 10,000+ records.
- Participated in system design discussions for scalable async workflows, stateless agent execution, and multi-user session isolation across regulatory AI services.

Tata Consultancy Services (TCS) Sep 2023 – Aug 2024

Site Reliability Engineer

- Executed SSO integration for LRE 23 supporting 1,000+ users, reducing authentication delays by 50%.
- Automated operational workflows via Python scripting, cutting manual intervention by 30% and improving deployment efficiency.
- Conducted RCA for production incidents and enforced HTTPS migration for ALM, reducing recurring issues by 35%.

Tata Consultancy Services (TCS) Mar 2021 – Sep 2023

DevOps Engineer

- Managed CI/CD pipelines via Bitbucket and TeamCity supporting agile releases for 50,000+ users.
- Enhanced SDLC tool stability (Jira, Confluence, SonarQube), increasing deployment reliability by 20% and reducing onboarding time by 30%.
- Improved incident detection by 40% through Grafana dashboards and enforced version control best practices across teams.

Projects

AI-Powered Financial News Intelligence Platform (RAG + Chatbot) Jan 2024 – Present

- Architected end-to-end GenAI platform ingesting 5+ data formats (RSS, PDFs, CSV, DOCX, HTML) using LangChain and FAISS-based RAG pipeline.
- Reduced query response time by 40% over keyword search by implementing semantic vector retrieval with FAISS across 10,000+ indexed financial documents.
- Built interactive chatbot supporting 10+ query types (sentiment, sector trends, price signals, market insights) with sub-3-second average response time.
- Consolidated multi-step LLM inference into a single optimized pipeline, cutting end-to-end processing latency by 35% and token usage by 25%.
- Deployed Streamlit dashboard visualizing real-time insights across 5+ financial metrics; ingestion pipeline processes 200+ new articles daily.

Gemini Multi-Turn Conversational Chatbot Personal Project

- Built multi-thread chatbot using LangGraph, Gemini 2.5 Flash, and SQLite checkpointing, supporting unlimited concurrent sessions with 100% context retention across restarts.
- Reduced session recovery time to under 1 second by optimizing SQLite checkpoint queries across threads storing full conversation state.
- Implemented LangGraph StateGraph with TypedDict state management, maintaining complete message history across 3+ conversation turns per session.
- Designed Streamlit interface with thread switching, real-time chat history, and new session creation; reduced UI interaction latency by 50% via session state caching.
- Achieved zero message loss on application restart across 100+ test sessions by leveraging SQLite-backed persistent checkpointing.

Education

B.Tech – Information Technology July 2020

Guru Nanak Dev Engineering College (GNDEC), Ludhiana CGPA: 7.96 / 10

Intermediate (BSEB) July 2016

MRJDI College, Begusarai Percentage: 75.6%

Certifications

Professional Data Engineer | **Professional Cloud Database Engineer** | **Associate Cloud Engineer** | **Generative AI Leader**

Credly Profile: View All Certifications

Awards & Recognition

TCS AI Idea Igniter Badge | **tcsAI Spark Badge** | **Applause Award** | **On The Spot Award** | **Service Commitment Award** | **TCS AI Fridays Season 2 Certificate of Achievement** (Apr 2026)